Nonlinear Lensing Likelihood and MeV Thermal Completion in WRRA

Working calculation v0.6 | Wonsik Choi and Jeongin Choi | Independent researchers | 20 October 2026

Verified inputs → WRRA transformation → outputs → falsification

Verified inputs. Retain the WRRA address-to-background dark_weight_only branch: H0=67.2568374662, omega_b=0.02239580679, omega_c=0.12037806177. Declare calibrated adiabatic As=2.1e-9, ns=0.9649, tau=0.0544, N_eff=3.044 and T_CMB=2.7255 K. Use the released 18-bin SPT-Planck covariance, windows and response corrections. The separate thermal module retains the v0.5 packet budget, N=256, 128 events and 1% record cost, and adds finite-mass equilibrium electron/positron thermodynamics. Known verified inputs are legitimate model construction inputs; their adoption does not constitute a new derivation.

WRRA transformation. Execute new CAMB transfer/lensing calculations with the WRRA-derived background and explicit primordial completion. Separately connect the residue-to-matter and record-heat event ledger directly to temperature, expansion and pressure work. Adopted external methods are distinguished from the connections actually executed in this calculation.

Outputs. The nonlinear fixed model gives chi-square=10.606013550 over 18 bins. Initial photon temperature is 1.75848508951 MeV; the entropy-connected physical initial scale is 9.55163454815e-11 when the old time unit is retained. All 54 checks pass; a clean folder repeat reproduces scientific JSON, CSV and arrays exactly.

Falsification. Reject covariance/normalization/response errors, failed official-code crosschecks, failed time/EOS refinement or conservation, and claims of complete state integration when the initial-state gate is blocked.

Assessment

The statistical and MeV thermal modules are implemented. Their existence is not completion of the entire Step 3 integration. The lensing result is conditional on the declared adiabatic completion. Full v0.4 metric/velocity/neutrino states and v0.5 event-time perturbations are not injected into these CAMB runs.

Released-data likelihood and new nonlinear spectra

Four new CAMB 2.0.4 runs compare linear lensing, HMCode 2020 nonlinear lensing, an inherited unweighted-background nonlinear control, and a 1.5-accuracy refinement. HMCode is an adopted empirical prescription, not a nonlinear law derived or validated specifically for the finite WRRA SOURCE.

$$D_L^{\phi\phi} = \frac{[L(L+1)]^2}{2\pi} C_L^{\phi\phi}, \qquad D_\ell^{TT} = \frac{\ell(\ell+1)}{2\pi} C_\ell^{TT} \quad [\mu\mathrm{K}^2],$$

$$b_i = \sum_L (W_{iL} + R_{iL}^{PP}) D_L^{\phi\phi} + \sum_\ell R_{i\ell}^{TT} D_\ell^{TT} - f_i,$$

$$\chi^2 = (b-d)^T \Sigma^{-1} (b-d), \qquad \log \mathcal{L} = -\frac{1}{2}[\chi^2 + \log|\Sigma| + 18\log(2\pi)].$$

Read W, R, f and Sigma from the released package. Use computed lensed TT in the TT response correction. Do not replace the covariance with plotting errors. All cosmological parameters in this comparison are fixed; no joint MCMC is run.

Case	chi-square, 18 bins	Profile A_phi	Profile sigma
linear	11.685583691	0.982136764	0.054517262
nonlinear	10.606013550	0.943444900	0.051928328
unweighted_nonlinear	10.545990268	0.958305978	0.054838355
nonlinear_refined	10.603227973	0.943426622	0.051918189

The profiled A_phi is a diagnostic lensing-amplitude fit at fixed shape, not a WRRA-derived constant or a new As calibration. Listed fixed-model chi-square values are before that fit. The background branches are too similar in this comparison to claim selection of one branch by these data alone.

The largest difference from the separate official CosmoMC implementation is 1.59872e-14. Only NumPy 2 type aliases are changed in memory for compatibility. The covariance-norm distance under accuracy refinement is 0.00383078 sigma. A zero-lensing control gives chi-square=339.503826141. That control verifies sensitivity to the observed lensing signal; it is not independent confirmation of the complete WRRA structure. A tail probability must not be interpreted as the probability that the theory is true.

Finite-packet thermal completion

At zero electron chemical potential, directly integrate the massive ideal Fermi gas. Refine quadrature and interpolation and independently differentiate the energy density to check heat capacity. Energy densities use natural MeV units; restore hbar to obtain H in inverse seconds.

$$\rho_{e^\pm} = \frac{4}{2\pi^2}\int_0^\infty \frac{p^2\sqrt{p^2+m_e^2}}{\exp(\sqrt{p^2+m_e^2}/T_\gamma)+1}\,dp,$$

$$p_{e^\pm} = \frac{4}{6\pi^2}\int_0^\infty \frac{p^4}{\sqrt{p^2+m_e^2}[\exp(\sqrt{p^2+m_e^2}/T_\gamma)+1]}\,dp.$$

$$\rho_\gamma = \pi^2 T_\gamma^4/15, \quad \rho_\nu = (7/8)(2N_{\text{eff}})\pi^2 T_\nu^4/30, \quad s_{\text{em}} = (\rho_{\text{em}}+p_{\text{em}})/T_\gamma,$$

$$H = \frac{\sqrt{8\pi\rho_{\text{tot}}/3}}{M_{\text{Pl}}\hbar}, \quad \dot{T}_\gamma = -\frac{3H(\rho_{\text{em}}+p_{\text{em}})}{d\rho_{\text{em}}/dT_\gamma}, \quad T_\nu \propto a^{-1}.$$

Use the relativistic instantaneous-decoupling entropy branch for the neutrino temperature ratio. N_eff=3.044 is an effective energy input, not a collision- integral result. At each event, add record heat to the electromagnetic energy density and solve for the new photon/electron temperature. Keep the paired reservoir debit and matter credit in the same event. This differs from simply adding an always-radiative J/a^4 term when electrons become nonrelativistic.

Photon T (MeV)	Elapsed seconds	Relative scale factor	Neutrino T (MeV)
1	0.504589442	1.76555227	0.994069083
0.5	2.82571512	3.59202382	0.488605035
0.1	118.621328	22.9948678	0.0763248972
0.05	528.985866	49.0916558	0.0357511046

Times are elapsed from the pulse start, not cosmic age. Keeping initial H=2.09590521352 /s gives T_initial=1.75848508951 MeV. Assuming no later non-standard photon entropy production, today's photon temperature implies a_initial=9.55163454815e-11, about 4.48365% below the old 1e-10. Thus the old clock, physical scale and particle content cannot all remain fixed while merely attaching a MeV label. This identifies a physical normalization revision.

Conservation, reproduction and integration boundary

Maximum total energy-plus-pressure-work error is 1.02816e-10. Reconstructing a small event heat increment from actual rounded temperatures gives a per-packet relative residual of 4.84538e-07. These use different normalizations; the latter includes subtractive roundoff of a tiny injection into a large thermal bath and is not the v0.5 algebraic ledger residual. The largest refinement change in the reported key thermal quantities is 1.47619e-10. Record heat increases electromagnetic comoving entropy. No complete microscopic total-universe entropy proof is claimed.

All 64 retained matched modes have nonzero matter velocity and 13 saved neutrino multipoles per mode. The current CAMB runs use adiabatic initial conditions, not these states. The complete-injection label is therefore blocked. Necessary subsequent work is concrete:

1. Transform the complete state to CAMB variables and the CDM reference frame.
2. Complete baryon/photon tight coupling and massive-neutrino states on the same background.
3. Propagate the thermal scale revision into finite k and conserved event-time junctions.
4. Fix the angular/statistical completion of the finite spatial seed, then inject all states.
5. Apply this likelihood to that result and extend to joint CMB/BAO/BBN posteriors.

Weak decoupling rates, flavor transport, the nuclear reaction network and BBN abundances are not computed here. Calibration with verified inputs is valid; calibration choices and outputs after fixing those choices must remain distinct.

Acknowledgments/references: NASA LAMBDA, SPT and Planck; Simard et al., arXiv:1712.07541; Omori et al., arXiv:1705.00743; CAMB; Mead et al., arXiv:2009.01858; Anthony Lewis's CosmoMC likelihood format; PDG 2025 Big-Bang Cosmology, https://pdg.lbl.gov/2025/reviews/rpp2025-rev-bbang-cosmology.pdf. Exact data URLs, code versions and SHA256 values are fixed in PROVENANCE.json.

Numerical diagnostics

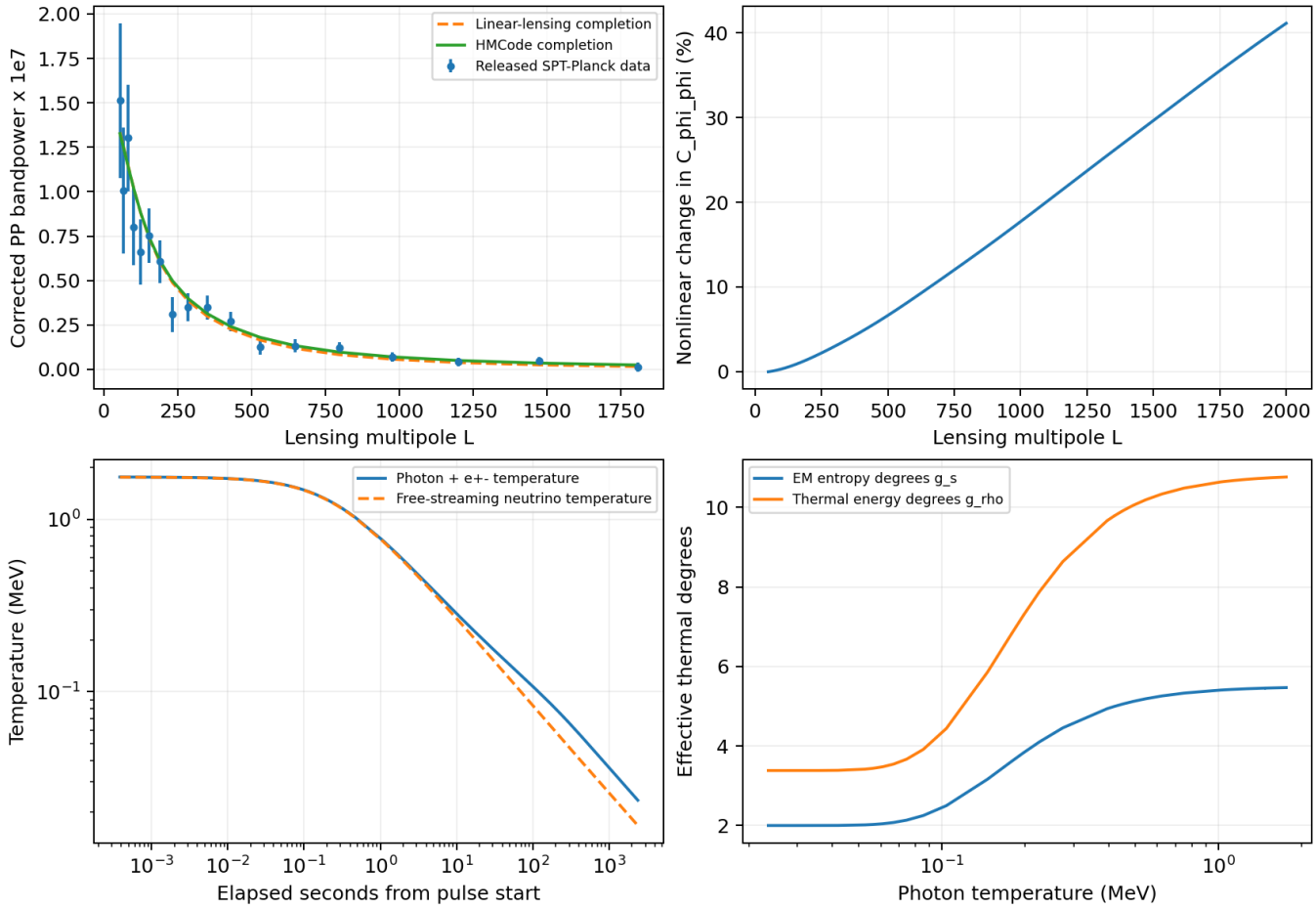


Figure 1: Released bandpowers and conditional predictions, nonlinear corrections, MeV temperature history and thermal degrees of freedom.

The lensing panel uses the WRRA background plus the calibrated adiabatic completion. Thermal history is a separate local event module. These figures do not assert a common full-state injection between the two modules.